Chapter 1

Integration in Several Variables

There are many aspects of integration in several variables. In this chapter we first discuss the definition and evaluation of the integral of well behaved *scalar functions* on simple domains in the Euclidean space $\mathbb{R}^n$ which are generalizations of rectangular boxes, triangles and tetrahedrons in two and three dimensions, called *hypercubes* and *simplexes* respectively. We then extend the discussion to *domains with piecewise C^1 boundary*. In the next chapter we will extend the discussion, where the domain of integration is a lower dimensional surface in the Euclidean space $\mathbb{R}^n$, and the integrand is given in terms of *coupling* of a vector field and the tangent plane of the surface, or more formally, the integration of a differential form over a surface or a **singular chain**.

Below are some main issues to be addressed.

1. It is not too different from integration in one dimension if one only discusses integrals of a scalar function defined in a multi-dimensional rectangle. One can establish a similar integrability criterion and prove a version of the **Fubini Theorem**, which reduces the evaluation of the integral to iterated one variable integrals.

2. Complications come in when one tries to define integrals in more general domains in multi-dimensions. One main issue is how to properly define **partitions**. Partitions in the Riemann sense require that there is a well defined notion of area (volume) for the *cells* used in the partition, which is why traditionally we only confine to rectangular cells in partitions. Another aspect is that we want the cells to be *non-overlapping*, and this is relatively easy to achieve using rectangular cells.

   One way to resolve the issue of decomposing a general domain as a non-overlapping union of rectangular cells is to enclose the domain (required to be bounded) by a rectangular box and extend the integrand to be 0 in the complement. This would require accounting for the contributions to the upper and lower integrals from those rectangular cells in the partition of the box that "*straddle*" between the domain and its complement. This is not too difficult to address, but one does need to control the *measure* of the boundary of the domain of integration.

3. In any integration theory one expects the integral $\int_A f$ to be linear in the

integrand f, namely, for any integrable f, g, and constants a, b,

$$\int_A (af + bg) = a \int_A f + b \int_A g.$$

It is also natural to expect that when A can be decomposed as a non-overlapping union of two subsets $A_1 \cup A_2$

$$\int_A f = \int_{A_1} f + \int_{A_2} f.$$

One also hopes that the converse holds: if the terms on the right hand side are well defined, then the left hand side should be well defined and equals the right hand side.

In one dimension this is certainly true when A_1, A_2 are intervals. But even in that context, Riemann integration theory places restrictions on these sets: if we take $A = [0, 1], A_1 = \mathbb{Q} \cap [0, 1], A_2 = [0, 1] \setminus A_1$, then it is reasonable to argue that Dirichlet's non-Riemann integrable function on $[0, 1]$ could be regarded as integrable on both A_1 and A_2 as it is a constant on either set, but its integral on $[0, 1]$ is not defined. One key issue is that Riemann's integration theory does not allow general sets such as A_2 here as cells in the partition, and each partition has to be a *finite* partition. As a result one can't directly define Riemann integral on a domain with infinite length (area or volume).

4. A more serious issue is that *reasonable limits of Riemann integrable functions may not be Riemann integrable.* The rectification of this issue requires by Lebesgue's integration theory.

5. The simplistic idea of using "nice cells" to partition a domain of integration becomes more challenging when one tries to carry it out to surfaces or hypersurfaces. The issue of defining the area (volume) of cells has to be tackled first.

6. A basic difficulty in discussing the integration on a surface is how to properly define a surface. Unlike in the case of a curve, which can always be thought of as defined through a map on an interval, there is no simple or canonical domain which can be used to define a surface, except for a *patch* of a surface.

7. For the purpose of integration theory on a surface, it suffices that one can partition the surface into some non-overlapping union of patches such that each patch has a *parametric representation* through a map defined on a nice domain such as a rectangular box or simplex, and one defines the integration on each such patch on this nice domain through this parametrization.

8. The implementation of the above approach still requires substantial preparation work. First, the existence and construction of such a partition would require careful proof. Second, there is usually no canonical parametrization for a patch of a surface, so one has to verify that different parametrizations do not lead to different values of the integrals.

1.1 Integral of a scalar function on multi-dimensional rectangles and more general domains

Definition 1.1.1 A set of the form $R = [a_1, b_1] \times \cdots \times [a_n, b_n]$, where $a_i < b_i$ for each i, is called a **hypercube** in $\mathbb{R}^n$. We also informally call it an n-dimensional rectangle (or rectangular box). We define its **volume** to be

$$|R| := (b_1 - a_1) \cdots (b_n - a_n).$$

When $a_i = b_i$ is allowed, and this equality holds for at least one i, we call such an R a degenerate hypercube (or informally a degenerate rectangle), and define its volume to be 0.

Suppose that $\mathcal{P}_i$ is a partition of $[a_i, b_i]$ for $1 \le i \le n$. Then these partitions $\mathcal{P}_i$'s form a partition $\mathcal{P}$ of $R = [a_1, b_1] \times \cdots \times [a_n, b_n]$ in the following sense:

(i). The cells of the partition $\mathcal{P}$ consist of $\{S := I_1 \times \cdots \times I_n : I_i \text{ is a subinterval of } \mathcal{P}_i\}$;

(ii). the union of the cells of $\mathcal{P} = I$;

(iii). the intersection of any two different cells of $\mathcal{P}$ is either the empty set, or a degenerate rectangle; and

(iv). $|R| = \sum_{S \in \mathcal{P}} |S|$.

We may denote this partition by $\mathcal{P}_1 \times \cdots \times \mathcal{P}_n$.

The *mesh size* $\lambda(\mathcal{P})$ of $\mathcal{P} := \mathcal{P}_1 \times \cdots \times \mathcal{P}_n$ is defined as

$$\lambda(\mathcal{P}) := \max\{|I_i| : S := I_1 \times \cdots \times I_n \in \mathcal{P}\}.$$

$\Diamond$

Remark 1.1.2 Only the verification of the last property requires more than a few lines of argument. One easy way is to use an induction argument on n, the dimension. $R' := [a_2, b_2] \times \cdots \times [a_n, b_n]$ is an $(n-1)$-dimensional rectangle, and the partitions $\{\mathcal{P}_2, \ldots, \mathcal{P}_n\}$ form a partition $\mathcal{P}'$ of R'. Suppose that $\{I_{1,1}, \ldots, I_{1,k}\}$ are the subintervals of $\mathcal{P}_1$. In the summation $\sum_{S \in \mathcal{P}} |S|$, group S according to its constitutive factor in the first variable, each S has the form of $I_{1,j} \times S'$ for some $I_{1,j} \in \mathcal{P}_1$ and $S' \in \mathcal{P}'$, thus we have

$$\sum_{S \in \mathcal{P}} |S| = \sum_{I_{1,j} \in \mathcal{P}_1} \sum_{S' \in \mathcal{P}'} |I_{1,j} \times S'|.$$

By induction, we have

$$\sum_{S' \in \mathcal{P}'} |I_{1,j} \times S'| = |I_{1,j}| \sum_{S' \in \mathcal{P}'} |S'| = |I_{1,j}||R'|.$$

It now follows that

$$\sum_{S \in \mathcal{P}} |S| = \sum_{I_{1,j} \in \mathcal{P}_1} |I_{1,j}||R'| = (b_1 - a_1)|R'| = |R|.$$

Note that a partition of R in our definition is defined in terms of a collection of partitions $\mathcal{P}_i$ for each constitutive factor. As a result, if the projection onto a factor of two rectangles S_1, S_2 in a partition of R share more than one points, they must have identical subinterval in that factor. One could allow more general choices of rectangles in a partition; it would just make it harder to bookkeep such rectangles, and our definition suffices for our purpose of defining the integral of a function.

Definition 1.1.3 Suppose that f is a bounded real-valued function defined on the rectangle R, $\mathcal{P}$ a partition of R, and a point $\mathbf{x}_\alpha \in R_\alpha$ is chosen for each sub rectangle R_α of $\mathcal{P}$. We define the corresponding **Riemann sum** as

$$R(f, \mathcal{P}, \{\mathbf{x}_\alpha\}) := \sum_\alpha f(\mathbf{x}_\alpha)|R_\alpha|.$$

Define

$$M_S(f) := \sup_{\mathbf{x}\in S} f(\mathbf{x}) \quad \text{and} \quad m_S(f) := \inf_{\mathbf{x}\in S} f(\mathbf{x})$$

as the supremum and, respectively, infimum of f on the rectangle S. Then the **upper sum** $U(f, \mathcal{P})$ and, respectively, the **lower sum** $L(f, \mathcal{P})$, of f on R with respect to the partition $\mathcal{P}$ is defined as

$$U(f, \mathcal{P}) := \sum_\alpha M_{S_\alpha}(f)|S_\alpha|, \quad L(f, \mathcal{P}) := \sum_\alpha m_{S_\alpha}(f)|S_\alpha|.$$

$\Diamond$

Note that

$$L(f, \mathcal{P}) \le R(f, \mathcal{P}, \{\mathbf{x}_\alpha\}) \le U(f, \mathcal{P}).$$

We are interested in whether $R(f, \mathcal{P}, \{\mathbf{x}_\alpha\})$ has a limit as $\lambda(\mathcal{P}) \to 0$ which is independent of how $\mathbf{x}_\alpha \in S_\alpha$ is chosen. It is easier to study whether $U(f, \mathcal{P})$and, respectively, $L(f, \mathcal{P})$, has a limit as $\lambda(\mathcal{P}) \to 0$.

Definition 1.1.4 A partition $\mathcal{P}^* := \mathcal{P}_1^* \times \cdots \times \mathcal{P}_n^*$ is called a **refinement** of partition $\mathcal{P} := \mathcal{P}_1 \times \cdots \times \mathcal{P}_n$ if each $\mathcal{P}_i^*$ is a refinement of $\mathcal{P}_i$. $\Diamond$

Note that if $\mathcal{P}^*$ is a refinement of partition $\mathcal{P}$, then $\lambda(\mathcal{P}^*) \le \lambda(\mathcal{P})$, and that

$$U(f, \mathcal{P}^*) \le U(f, \mathcal{P}), L(f, \mathcal{P}^*) \ge L(f, \mathcal{P}).$$

Just as in one dimension, if $\mathcal{P}_1, \mathcal{P}_2$ are two partitions of R, then there is a partition $\mathcal{P}^*$ which is a refinement of both $\mathcal{P}_1$ and $\mathcal{P}_2$, and there is an essentially canonical way of constructing such an $\mathcal{P}^*$ by adjoining all the end points of the subintervals of the factors of $\mathcal{P}_1, \mathcal{P}_2$.

Proposition 1.1.5 Upper and lower integral of a bounded function defined on a rectangle. *Suppose that f is a bounded function defined on the rectangle $R \subset \mathbb{R}^n$. Then*

$$\inf_{\mathcal{P}} U(f, \mathcal{P}) = \lim_{\lambda(\mathcal{P})\to 0} U(f, \mathcal{P}),$$

and

$$\sup_{\mathcal{P}} L(f, \mathcal{P}) = \lim_{\lambda(\mathcal{P})\to 0} L(f, \mathcal{P}).$$

Furthermore,

$$\sup_{\mathcal{P}} L(f, \mathcal{P}) \le \inf_{\mathcal{P}} U(f, \mathcal{P}).$$

Proof. Let $\mathcal{P}_1, \mathcal{P}_2$ be two arbitrary partitions of R, and $\mathcal{P}^*$ be a refinement of both $\mathcal{P}_1$ and $\mathcal{P}_2$. Then

$$L(f, \mathcal{P}_1) \leq L(f, \mathcal{P}^*) \leq U(f, \mathcal{P}^*) \leq U(f, \mathcal{P}_2).$$

As a result, $\bar{\int}_R f$, $\underline{\int}_R f$

$$L(f, \mathcal{P}_1) \leq \bar{\int}_R f := \inf_{\mathcal{P}_2} U(f, \mathcal{P}_2),$$

and

$$\underline{\int}_R f := \sup_{\mathcal{P}_1} L(f, \mathcal{P}_1) \leq \bar{\int}_R f.$$

For any $\epsilon > 0$, first find a partition $\mathcal{P}_2$ of R such that

$$\bar{\int}_R f \leq U(f, \mathcal{P}_2) \leq \bar{\int}_R f + \epsilon.$$

Now take any partition $\mathcal{P}$ of R of small mesh size $\lambda(\mathcal{P})$ (to be determined). Let $\mathcal{P}^*$ be the canonical refinement of $\mathcal{P}$ and $\mathcal{P}_2$. Then there exists a number N depending only on the partition $\mathcal{P}_2$ such that in the sums $U(f, \mathcal{P})$ and $U(f, \mathcal{P}^*)$, all but at most N terms may differ, and each term that may differ is bounded above in absolute value by $\sup_R |f|(\lambda(\mathcal{P}))^n$, thus

$$|U(f, \mathcal{P}) - U(f, \mathcal{P}^*)| \leq N \sup_R |f|(\lambda(\mathcal{P}))^n.$$

It now follows that there exists some $\delta > 0$ such that whenever $\lambda(\mathcal{P}) < \delta$ we have

$$|U(f, \mathcal{P}) - U(f, \mathcal{P}^*)| < \epsilon \text{ so } U(f, \mathcal{P}) < \bar{\int}_R f + 2\epsilon,$$

proving that $\lim_{\lambda(\mathcal{P}) \to 0} U(f, \mathcal{P}) = \bar{\int}_R f$. ■